# Type MS · Merge sort: split, then merge

## Answer 1 of 2 · Question 2 part c · 5 marks

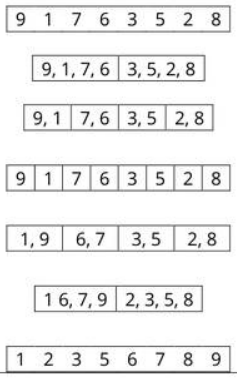
2021 NOV · Paper 02 · extract 1 of 18 · find part c in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (b) | B1<br>B2<br>B3<br>B4 | **Award up to 4 marks for a linked explanation such as:**<br><br>• The list is not sorted (1) Juan should come after Elija if the list was sorted / Elija is compared with Juan (1), so the bottom half of the list would be discarded (1) after the first pass through the loop (1)<br>• Binary search works on a sorted list (1) because it uses divide and conquer (1) half the list discarded each pass through (1) based on comparison of search item with middle item (1) | | (4) |
| **Question** | **mp** | **Answer** | **Additional Guidance** | **Mark** |

# Type MS · Merge sort: split, then merge

## Answer 1 of 2 · Question 2 part c · 5 marks

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| | | | | |
|---|---|---|---|---|
| 2 (c) | C1<br>C2<br>C3<br>C4<br>C5 | **Award 1 mark for each of:**<br>Split into two sets of four (1)<br>Split each set into two pairs (1)<br>Split each pair into single elements (1)<br>Merge elements into sorted pairs (1)<br>Merge pairs into sorted sets (1)<br>  | Allow step at C3 to be implied | (5) |

# Type MS · Merge sort: split, then merge

## Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 3 of 18 · find part c in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Merge sort: split, then merge

### Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 4 of 18 · find part c in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a) | **Award one mark for each of:** | | Logic of algorithm must be followed as set out.<br>Alternatives must address each point.<br>Do not penalise candidates who attempt more than the stated requirements.<br>Do not penalise spelling mistakes in the input message. | |
| | A1 | Input as an integer (1) | | |
| | A2 | Loop with correct conditions<br>start <= end and not found (1) | | |
| | A3 | Use of correct symbol for integer division (in conversion of DIV) or type coercion (1) | | |
| | A4 | Check to see if number in middle matches item (1) | | |
| | A5 | Set found to true (1) | | |
| | A6 | Check to see if item is less than number in middle (1) | | |
| | A7 | Set end to middle – 1 (1) | | |
| | A8 | else – set start to middle + 1 (1) | | |
| | A9 | count incremented by 1 (1) | | |
| | A10 | Program execution is fully correct (1) | | (10) |
| Code examples | | | | |

## Type MS · Merge sort: split, then merge

### Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 5 of 18 · find part c in this mark scheme extract

C#

```
Console.WriteLine("What is the number to find? ");
int item = Convert.ToInt32(Console.ReadLine());

while (start <= end && !found)
{
    middle = (start + end) / 2;
    if (numberList[middle] == item)
    {
        found = true;
    }
    else
    {
        if (item < numberList[middle])
        {
            end = middle - 1;
        }
        else
        {
            start = middle + 1;
        }
    }
    count++;
}
```

## Type MS · Merge sort: split, then merge

### Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 6 of 18 · find part c in this mark scheme extract

Java

```
Scanner input = new Scanner(System.in);
System.out.print("What is the number to find? ");
int item = Integer.parseInt(input.nextLine());

while (start <= end && !found)
{
    middle = (start + end) / 2;
    if (numberList[middle] == item)
    {
        found = true;
    }
    else
    {
      if (item < numberList[middle])
      {
          end = middle - 1;
      }
      else
      {
          start = middle + 1;
      }
    }
    count ++;
}
```

Python

```
item = int(input("What is the number to find? "))
while start <= end and not found:
    middle = (start + end) // 2
    if numberList[middle] == item:
        found = True
    else:
        if item < numberList[middle]:
            end = middle - 1
        else:
            start = middle + 1
    count = count + 1
```

## Type MS · Merge sort: split, then merge

### Answer 1 of 2 · Question 2 part c · 5 marks

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| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Merge sort: split, then merge

### Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 8 of 18 · find part c in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 4 (a) | **Award one mark for each of:** | | | |
| | | **Validation** | | |
| | A1 | Suitable input prompt and storing it in binaryPattern (1) | | |
| | A2 | Length of binary pattern validated as 8 (1) | | |
| | A3 | Loop used to check the length of binary pattern (1) | | |
| | | **Binary to denary** | | |
| | A4 | Loop through each digit in the binary pattern (1) | | |
| | A5 | Correct conversion of at least one binary digit to relevant placeholder (1) | | |
| | A6 | Attempt at running total using denaryNumber (1) | | |
| | A7 | Correct running total using denaryNumber (1) | | |
| | A8 | Print statement includes the binary pattern and the denary number (1) | | |
| | A9 | Print statement outside of conversion loop (1) | | |
| | A10 | Comment explains how conversion works (1) | | |
| | A11 | Program executes correctly for any binary pattern that is exactly 8 characters long only (1) | | (11) |
| Code examples | | | | |

## Type MS · Merge sort: split, then merge

### Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 9 of 18 · find part c in this mark scheme extract

C#

```
while (binaryPattern.Length != 8)
{
    Console.WriteLine("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = Console.ReadLine();
}

foreach (char digit in binaryPattern)
{
    if (digit == '1')
    {
        denaryNumber += denaryPlaceholders[count] * 1;
    }
    count++;
}
Console.WriteLine(binaryPattern + " converted to denary is " + denaryNumber);
Console.ReadLine();
```

Java

```
while (binaryPattern.length() != 8)
{
    Scanner input = new Scanner(System.in);
    System.out.print("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = input.next();
}

for (char digit : binaryPattern.toCharArray())
{
    if (digit == '1')
    {
       denaryNumber += denaryPlaceholders[count] * 1;
    }
    count ++;
}
System.out.print(binaryPattern + " converted to denary is " + denaryNumber);
```

## Type MS · Merge sort: split, then merge

### Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 10 of 18 · find part c in this mark scheme extract

Python

```
while len(binaryPattern) != 8:
    binaryPattern = input("Enter an 8 digit binary number to convert to denary: ")

for digit in binaryPattern:
    if digit == "1":
        denaryNumber = denaryNumber + denaryPlaceholders[count] * 1
    count = count + 1

print(binaryPattern,"converted to denary is", denaryNumber)
```

# Type MS · Merge sort: split, then merge

## Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 11 of 18 · find part c in this mark scheme extract

For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 5 | A1 | At least three variables initialised appropriately | | (1) |
| | A2 | Word input | | (1) |
| | A3 | Program repeats until 1 is input | | (1) |
| | A4 | Loop through each word in the array | | (1) |
| | | Words that begin with the same letter | | |
| | A5 | Check to see if the first letter of the input word matches the first letter of at least one word in the array | | (1) |
| | A6 | The number of words that begin with the same letter displayed | | (1) |
| | | Words that contain the input word | | |
| | A7 | Check to see if the word contains the input word | | (1) |
| | A8 | Number of letters in the word calculated | | (1) |
| | A9 | At least one word from the array that contains the input word displayed | | (1) |
| | A10 | The number of words that contain the input word displayed | | (1) |
| | A11 | Identification of the longest word or the shortest word | | (1) |

| Band 1 (1-3 marks) | Band 2 (4-6 marks) | Band 3 (7-9 marks) | Mark |
|---|---|---|---|
| Little attempt to decompose into component parts | Some attempt to decompose into component parts | The problem has been decomposed into component parts | |
| Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | (9) |

# Type MS · Merge sort: split, then merge

## Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 12 of 18 · find part c in this mark scheme extract

Code examples

C#

```
// Add your code here
string inputWord = "";

while (inputWord != "1")
{
    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    Console.WriteLine("Enter a word or 1 to exit ");
    inputWord = Console.ReadLine();

    if (inputWord != "1")
    {
        int count = 0;

        foreach (String word in wordArray)
        {
            if(word[0] == inputWord[0])
            {
                Console.WriteLine(word);
                count ++;
            }
        } // End of Loop

        Console.WriteLine(count + " word(s) beginning with "+ inputWord[0]);
        Console.WriteLine("------------------------------------------------------------------------");

        count = 0;

        foreach (String word in wordArray)
        {
            if(word.Contains(inputWord))
            {
                count ++;
                Console.WriteLine(word);

                int length = word.Length;

                if (longest < length)
```

# Type MS · Merge sort: split, then merge

## Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 13 of 18 · find part c in this mark scheme extract

```
                    {
                        longest = length;
                        longestWord = word;
                    }

                    if (shortest > length)
                    {
                        shortest = length;
                        shortestWord = word;
                    }
                }// end of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word

            if (count > 0)
            {
                Console.WriteLine(count + " word(s) with "+ inputWord + " in them");
                Console.WriteLine("The longest word has " + longest + " characters");
                Console.WriteLine("The shortest word has " + shortest + " characters");
                Console.WriteLine("The longest word is " + longestWord + " characters");
                Console.WriteLine("The shortest word is " + shortestWord + " characters");
                Console.WriteLine("--------------------------------------------------------------------------------");
            }
            else
            {
                Console.WriteLine("There were 0 words that had all the letters from " + inputWord + " in them");
                Console.WriteLine("--------------------------------------------------------------------------------");
            } // End of printing words
        } // End of check
    } // End of while loop
```

# Type MS · Merge sort: split, then merge

## Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 14 of 18 · find part c in this mark scheme extract

Java

```
// Add your code here
Scanner input = new Scanner(System.in);
String inputWord = "";

// While there is a word run the program
while (!inputWord.equals("1")) {

    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    System.out.println("Enter a word or 1 to exit ");
    System.out.println("--------------------------------------------------------------------------------");
    inputWord = input.nextLine();
    if (!inputWord.equals("1")){
        // Find words that begin with the same letter as the input word
        int count = 0;

        // Get each word in the array
        for (String word: wordArray){

            // If the first character of the word is the same as the first character of
            // the input word then print the word
            if (word.charAt(0) == inputWord.charAt(0)){
                System.out.println(word);
                count ++;
            }
        } // End of for loop

        // Print the total number of words that begin with the same letter
        System.out.println(count + " word(s) beginning with "+ inputWord.charAt(0));
        System.out.println("--------------------------------------------------------------------------------");

        // Find words that contain the input word
        count = 0;

        // Get each word in the array
        for (String word: wordArray){
            // Check to see if the word contains the inputWord
            if (word.contains(inputWord)){
                count ++;
                System.out.println(word);

                // Set the length of the word
                int length = word.length();
```

# Type MS · Merge sort: split, then merge

## Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 15 of 18 · find part c in this mark scheme extract



```
                    // If the length is longer than the current longest replace it and the longestWord
                    if (longest < length) {
                        longest = length;
                        longestWord = word;
                    }

                    // If the length is shorter than the shortest replace it and the shortestWord
                    if (shortest > length){
                        shortest = length;
                        shortestWord = word;
                    }
                } // End of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word
            if( count > 0){
                System.out.println(count + " word(s) with "+ inputWord + " in them");
                System.out.println("The longest word has " + longest + " characters");
                System.out.println("The shortest word has " + shortest + " characters");
                System.out.println("The longest word is " + longestWord + " characters");
                System.out.println("The shortest word is " + shortestWord + " characters");
                System.out.println("--------------------------------------------------------------------------------");
            } else {
                System.out.println("There were 0 words that had all the letters from " + inputWord + " in them");
                System.out.println("--------------------------------------------------------------------------------");
            } // End of printing words
        }
    } // End of while loop
```

# Type MS · Merge sort: split, then merge

## Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 16 of 18 · find part c in this mark scheme extract

Python

```
# While there is a word run the program
while inputWord !="1":

    shortest = 5000
    longest = 0
    shortestWord = ""
    longestWord = ""

    inputWord = input("Enter a word or 1 to exit: ")
    print("------------------------------------------------------------------------")

    if inputWord !="1":
        # Find words that begin with the same letter as the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # If the first character of the word is the same as the first character of
            # the input word then print the word
            if word[0] == inputWord[0]:
                print(word)
                count = count + 1

        # Print the total number of words that begin with the same letter
        print("\n",count,"word(s) beginning with",inputWord[0])
        print("------------------------------------------------------------------------")

        # Find words that contain the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # Check to see if the word contains the inputWord
            if inputWord in word:
                count = count + 1
                print(word)
                # Set the length of the word
                length = len(word)

                # If the length is longer than the current longest replace it and the longestWord
                if longest < length:
                    longest = length
                    longestWord = word

                # If the length is shorter than the shortest replace it and the shortestWord
                if shortest > length:
                    shortest = length
                    shortestWord = word
```

# Type MS · Merge sort: split, then merge

## Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 17 of 18 · find part c in this mark scheme extract

```
59
60          # Print the number of words, the number of characters in the longest and
61          # shortest word, the longest and the shortest word
62          if count > 0:
63              print("\n",count,"word(s) with",inputWord,"in them")
64              print("The longest word has",longest,"letters")
65              print("The shortest word has",shortest,"letters")
66              print("The longest word is",longestWord)
67              print("The shortest word is",shortestWord)
68              print("-------------------------------------------------------------------------")
69          else:
70              print("There were 0 words with all the letters from",inputWord,"in them")
71          # ------------------------------------------------------------------------------------
72
73  # End the program when there are no more words
74  print("End of program")
```

## Type MS · Merge sort: split, then merge

### Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 18 of 18 · find part c in this mark scheme extract

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# Type MS · Merge sort: split, then merge

## Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 1 of 9 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 5(a) | A1<br>A2 | Award 1 mark for demonstration of one correctly merged/split row<br>Award 2 marks for correct merge sort<br><br>Blackfin, Bigeye \| Longtail, Albacore \| Bluefin<br><br>Longtail, Blackfin, Bigeye, Albacore \| Bluefin<br><br>Longtail, Bluefin, Blackfin, Bigeye, Albacore<br><br>OR<br><br>Bigeye \| Blackfin, Albacore \| Longtail, Bluefin<br><br>Bigeye \| Longtail, Bluefin, Blackfin, Albacore<br><br>Longtail, Bluefin, Blackfin, Bigeye, Albacore<br><br>OR<br><br>Blackfin, Bigeye \| Albacore \| Longtail, Bluefin<br><br>Blackfin, Bigeye \| Longtail, Bluefin, Albacore | We are testing pupils understanding of how a merge sort progresses:<br>• Divide and conquer (split)<br>• Merging adjacent elements<br>Allow answers in ascending order | **2** |

## Type MS · Merge sort: split, then merge

### Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 2 of 9 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | | Longtail, Bluefin, Blackfin, Bigeye, Albacore<br><br>OR<br><br>Blackfin, Bigeye \| Albacore \| Longtail, Bluefin<br><br>Blackfin, Bigeye, Albacore \| Longtail, Bluefin<br><br>Longtail, Bluefin, Blackfin, Bigeye, Albacore | | |

## Type MS · Merge sort: split, then merge

### Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 3 of 9 · find part a in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(b)</td><td>B1<br>B2</td><td>Award one mark for each correct cell:<br><table><tr><td>Line number with error</td><td>11 (1)</td></tr><tr><td>Corrected line of pseudocode</td><td>SET tmp TO myTuna[ndx] (1)</td></tr></table></td><td>Ignore spellings<br><br>Ignore missing text as long as the minimum of [ndx] is provided<br><br>B2 does not depend on B1</td><td>2</td></tr>
</table>

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 5(c) | Award **one** mark for each of: | | C1 allow append as writing to file<br><br>C4 Do not award comma within an array<br><br>C5 must have 1 complete line written to file (allow additional spaces and commas). Should include individual elements (2D index) and not a 1D array from tbl_tuna<br><br>C6 complete file must be as expected e.g. no additional comma at end of line and no spaces around commas | **6** |

## Type MS · Merge sort: split, then merge

### Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 4 of 9 · find part a in this mark scheme extract

**C#**

```
// Q05cFINISHED

using System.IO;

String[,] tblTuna =  {
    { "Yellowfin", "105", "15", "3"},
    { "Albacore","90","15","5"},
    { "Skipjack","50","3","4"},
    { "Bigeye","105","25","4"},
    { "Atlantic Bonito","50","4","2"},
    { "Northern Bluefin","190","120","11"},
    { "Southern Bluefin","190","120","11"},
    { "Tongol","90","20","4"}
};



// ------------------------------------------
// Write your code below this line

String fileName = "TunaData.txt";       // Output file name
String comma = ",";                     // Use as a constant
String full_path = "C:\\Q05c";
fileName = full_path + "\\" + fileName;

// Create variables as needed
int number = 101;           // Number for the line at the front
String lineOut = "";        // The line to output

// Get the dimensions of the array for looping
int rows = tblTuna.GetLength(0);
int columns = tblTuna.GetLength(1);

// Open the file for writing
StreamWriter fileWriter = new StreamWriter(fileName);

```

## Type MS · Merge sort: split, then merge

### Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 5 of 9 · find part a in this mark scheme extract

```
// Process every tuna in the table
for (int i = 0; i < rows; i++)          // Rows
{
    // Start with the number and a comma
    lineOut = number.ToString() + comma;

    for (int j = 0; j < columns; j++)   // Columns
    {
        // Add the name and the numbers
        lineOut = lineOut + tblTuna[i, j];

        // Add a comma to all the fields except the last
        if (j < columns - 1)
        {
            lineOut = lineOut + comma;
        }
    }
    // Write the line to the file
    fileWriter.WriteLine(lineOut);

    // Go to the next number for the line
    number = number + 1;
}

// Close the file
fileWriter.Close();
```

Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 6 of 9 · find part a in this mark scheme extract

**Java**

```java
// Q05cFINISHED

import java.io.FileWriter;
import java.io.IOException;

public class Q05cFINISHED {

    public static void main(String[] args) throws Exception {

        String[][] tblTuna =
        {
            { "Yellowfin", "105", "15", "3"},
            { "Albacore","90","15","5"},
            { "Skipjack","50","3","4"},
            { "Bigeye","105","25","4"},
            { "Atlantic Bonito","50","4","2"},
            { "Northern Bluefin","190","120","11"},
            { "Southern Bluefin","190","120","11"},
            { "Tongol","90","20","4"}
        };

        // ------------------------------------------
        // Write your code below this
        String fileName = "TunaData.txt";       // Output file name
        String comma = ",";                     // Use as a constant
        String full_path = "C:\\src\\q05c";
        fileName = full_path + "\\" + fileName;

        // Create variables as needed
        int number = 101;           // Number for the line at the front
        String lineOut = "";        // The line to output
        String linefeed = "\n";     // Use as a constant

        // Get the dimensions of the array for looping
        int rows = tblTuna.length;
        int columns = tblTuna[0].length;

        // Open the file for writing
        FileWriter fileWriter = new FileWriter(fileName);

```

## Type MS · Merge sort: split, then merge

### Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 7 of 9 · find part a in this mark scheme extract

```
41          // Process every tuna in the table
42          for (int i = 0; i < rows; i++)          // Rows
43          {
44              // Start with the number and a comma
45              lineOut = Integer.toString(number) + comma;
46
47              for (int j = 0; j < columns; j++)   // Columns
48              {
49                  // Add the name and the numbers
50                  lineOut = lineOut + tblTuna[i][j];
51
52                  // Add a comma to all the fields except the last
53                  if (j < columns - 1)
54                  {
55                      lineOut = lineOut + comma;
56                  }
57                  else
58                  {
59                      lineOut = lineOut + linefeed;
60                  }
61              }
62              // Write the line to the file
63              fileWriter.write(lineOut);
64
65              // Go to the next number for the line
66              number = number + 1;
67          }
68
69          // Close the file
70          fileWriter.close();
71      }
72
73  }
```

Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 8 of 9 · find part a in this mark scheme extract

**Python**

```
# Q05cFINISHED

tbl_tuna = [["Yellowfin",105,15,3],
            ["Albacore",90,15,5],
            ["Skipjack",50,3,4],
            ["Bigeye",105,25,4],
            ["Atlantic Bonito",50,4,2],
            ["Northern Bluefin",190,120,11],
            ["Southern Bluefin",190,120,11],
            ["Tongol",90,20,4]]

# ---------------------------------------------
# Write your code below this line

FILE_OUT = "TunaData.txt"           # Output file name
COMMA = ","                         # Use as constant

# Create variables as needed
LINEFEED = "\n"
number = 101                        # Number at the front
line_out = ""                       # The line to write

# Open the file for writing
file = open (FILE_OUT, "w")

# Process every tuna in the table
for tuna in tbl_tuna:
    # Start with the number and a comma
    line_out = str (number) + COMMA

    # Add the name of the tuna and a comma
    line_out = line_out + tuna[0] + COMMA

    # Each of the numbers need to be converted to a string
    # before adding to the output string.
    line_out = line_out + str (tuna[1]) + COMMA
    line_out = line_out + str (tuna[2]) + COMMA
    line_out = line_out + str (tuna[3])     # No comma on last field

    # Add a line feed to the whole line
    line_out = line_out + LINEFEED

    # Write the line to the file
    file.write (line_out)

    # Go to the next number for the line
    number = number + 1

# Close the file
file.close ()
```

## Type MS · Merge sort: split, then merge

### Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 9 of 9 · find part a in this mark scheme extract

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|